

C PROGRAMMING LAB REPORT

Area of Combined Geometric Figures

Question 1: Rectangle with a Semicircular End

Step 1 - Understand the Problem

Find the total area of a figure consisting of a rectangle and one semicircular end.

Item	Details
Inputs	Rectangle length (L), rectangle breadth (B), semicircle radius (R)
Output	Total area of the combined figure
Formula	Rectangle area = $L \times B$; Semicircle area = $0.5 \times \pi \times R \times R$; Total area = Rectangle area + Semicircle area

Step 2 - Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	L=10, B=4, R=2	46.28	46.28	Pass
2	L=8, B=5, R=3	54.14	54.14	Pass
3	L=12.5, B=6, R=1.5	78.53	78.53	Pass

Step 3 - Algorithm and Evaluation

1. Start
2. Declare L, B, R, rectangleArea, semicircleArea and totalArea as real numbers.
3. Read L, B and R.
4. Calculate $\text{rectangleArea} = L \times B$.
5. Calculate $\text{semicircleArea} = 0.5 \times \pi \times R \times R$.
6. Calculate $\text{totalArea} = \text{rectangleArea} + \text{semicircleArea}$.
7. Display totalArea.
8. Stop

Algorithm evaluation: The algorithm is finite, unambiguous, and directly applies the required geometric formulas. For the three prepared test cases, the calculated actual output matches the expected output to two decimal places.

Step 4 - C Program

```
#include <stdio.h>
#define PI 3.14159

int main() {
    double length, breadth, radius, rectangleArea, semicircleArea, totalArea;
```

```

printf("Enter rectangle length, rectangle breadth and semicircle radius: ");
scanf("%lf %lf %lf", &length, &breadth, &radius);

rectangleArea = length * breadth;
semicircleArea = 0.5 * PI * radius * radius;
totalArea = rectangleArea + semicircleArea;

printf("Total area = %.2f square units\n", totalArea);
return 0;
}

```

Step 5 - Compile, Execute and Evaluate

Compilation was performed with GCC. The program compiled successfully with no errors. It was then executed for each test case in Step 2. The actual outputs are shown in the test table and all test cases passed.

Step 6 - Compilation and Execution Evidence

```

$ gcc q1.c -o q1
Compilation successful.

$ ./q1
10 4 2
Enter rectangle length, rectangle breadth and semicircle radius: Total area = 46.28 square units

$ ./q1
8 5 3
Enter rectangle length, rectangle breadth and semicircle radius: Total area = 54.14 square units

$ ./q1
12.5 6 1.5
Enter rectangle length, rectangle breadth and semicircle radius: Total area = 78.53 square units

```

Figure: GCC compilation and execution results

Question 2: Triangle and Rectangle Combined Figure

Step 1 - Understand the Problem

Find the total area of a combined figure made of one triangle and one rectangle.

Item	Details
Inputs	Triangle base (b), triangle height (h), rectangle length (L), rectangle breadth (B)
Output	Total area of the combined figure
Formula	Triangle area = $0.5 \times b \times h$; Rectangle area = $L \times B$; Total area = Triangle area + Rectangle area

Step 2 - Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	b=6, h=4, L=10, B=5	62.00	62.00	Pass
2	b=8, h=5, L=7, B=3	41.00	41.00	Pass
3	b=5.5, h=3, L=4.5, B=2	17.25	17.25	Pass

Step 3 - Algorithm and Evaluation

1. Start
2. Declare b, h, L, B, triangleArea, rectangleArea and totalArea as real numbers.
3. Read b, h, L and B.
4. Calculate $\text{triangleArea} = 0.5 \times b \times h$.
5. Calculate $\text{rectangleArea} = L \times B$.
6. Calculate $\text{totalArea} = \text{triangleArea} + \text{rectangleArea}$.
7. Display totalArea.
8. Stop

Algorithm evaluation: The algorithm correctly separates the two component areas and then adds them. All three prepared test cases produced actual outputs equal to the expected outputs, so the program passes the test set.

Step 4 - C Program

```
#include <stdio.h>

int main() {
    double base, height, length, breadth, triangleArea, rectangleArea, totalArea;
```

```

    printf("Enter triangle base, triangle height, rectangle length and rectangle breadth:
");
    scanf("%lf %lf %lf %lf", &base, &height, &length, &breadth);

    triangleArea = 0.5 * base * height;
    rectangleArea = length * breadth;
    totalArea = triangleArea + rectangleArea;

    printf("Total area = %.2f square units\n", totalArea);
    return 0;
}

```

Step 5 - Compile, Execute and Evaluate

Compilation was performed with GCC. The program compiled successfully with no errors. It was then executed for each test case in Step 2. The actual outputs are shown in the test table and all test cases passed.

Step 6 - Compilation and Execution Evidence

```

$ gcc q2.c -o q2
Compilation successful.

$ ./q2
6 4 10 5
Enter triangle base, triangle height, rectangle length and rectangle breadth: Total area = 62.00 square units

$ ./q2
8 5 7 3
Enter triangle base, triangle height, rectangle length and rectangle breadth: Total area = 41.00 square units

$ ./q2
5.5 3 4.5 2
Enter triangle base, triangle height, rectangle length and rectangle breadth: Total area = 17.25 square units

```

Figure: GCC compilation and execution results

Final Conclusion

Both C programs were compiled and executed successfully. For every prepared test case, the actual result matched the expected result. Therefore, both programs satisfy the stated requirements for calculating the total area of their respective combined figures.